🏗️ The Scenario: Aerotech Logistics Optimisation

You have been hired as a Data Analyst for **Aerotech Engineering**, a company that manufactures precision aerospace components and manages time-critical freight. The engineering team is facing major supply chain inefficiencies due to departure delays.

Your task is to analyze performance metrics to optimize operations, improve predictive maintenance schedules, and evaluate carrier reliability.

❓ Step 2: The 3 Engineering Challenges

You need to write an R script to answer these three specific operational questions:

- **Problem 1 (Data Engineering & Cleaning):** Build a clean dataframe named `aerotech_clean`. You must exclude all cancelled or incomplete records (drop rows where `dep_time`, `arr_time`, or `air_time` are missing). Add a new column called `speed_mph` calculating the plane's average speed in miles per hour (`distance` divided by `air_time` converted to hours). Keep only the flights that traveled **above 400 miles**.
- **Problem 2 (Quality Engineering & Reliability):** Group the clean data by `carrier` to isolate the **top 5 worst-performing carriers** based on their average departure delay (`dep_delay`). For these 5 carriers, display their average delay, total flight count, and maximum recorded delay.
- **Problem 3 (Process Optimization & Bottlenecks):** Group the data by the hour of departure (`hour`). Calculate the average departure delay for each hour, and sort the data from the highest delay to the lowest to pinpoint exactly when ground crews are struggling the most.